

# C Programming Exam: Contact Management System

Total Points: 20

Duration: 90 minutes

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## □ Instructions

- Use **standard C libraries** only.
- Use **pointers** appropriately where required.
- Validate inputs and handle edge cases clearly.
- Code must be modular, clear, and well-commented.

## Question 1: Define Structure and Menu (3 Points)

### Task:

- Define a **Contact** structure with fields:
  1. **name**: string (max 50 chars)
  2. **phone**: string (max 15 chars)
  3. **age**: integer
- Display a menu with options:
  1. Add Contact
  2. View All Contacts
  3. Search by Name
  4. Delete by Phone
  5. Exit

### Points Distribution:

- Structure definition: 1 Point

- Menu loop and correct display: 2 Points

## Question 2: Add and Display Contacts (5 Points)

### Task:

- Use a static array to store up to **10 contacts**.
- Validate:
  - Name must not be empty
  - Phone number must be unique
  - Age must be  $> 0$
- Use a **pointer to add new contacts** into the array (e.g., pass pointer to the array and pointer to count).
- Display all contacts in tabular format using pointer-based iteration.

### Points Distribution:

- Add with validation using pointer logic: 3 Points
- Display using pointer traversal: 2 Points

## Question 3: Search by Name (4 Points)

### Task:

- Prompt user to enter a name (or partial name).
- Use a pointer-based function to **search for matches** (case-insensitive).
- Display all matches, or a message if none are found.

### Points Distribution:

- Pointer-based search logic: 2 Points
- Correct display of result or “not found” message: 2 Points

## Question 4: Delete by Phone Number (5 Points)

**Task:**

- Prompt for phone number to delete.
- Use a pointer to locate the record.
- If found, shift the remaining contacts to overwrite the deleted one.
- Display success or error message.

**Points Distribution:**

- Locate using pointer logic: 2 Points
- Deletion and shifting via pointer operations: 2 Points
- Success/failure message: 1 Point

**Question 5: Input Validation and Error Handling (3 Points)****Task:**

Ensure that:

- No contact is added if the list is full.
- Invalid menu options are handled gracefully.
- Clear and user-friendly error messages are shown.

**Points Distribution:**

- Full list detection and handling: 1 Point
- Invalid menu choice handling: 1 Point
- Proper prompts and error messages: 1 Point

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**☐ Submission Instructions**

- Submit `.c` source code files.
- Ensure your code compiles and runs correctly.

- Add explanatory comments in each function.
- Zip and name the submission folder as:  
`C_ContactManager_<YourName>.zip`